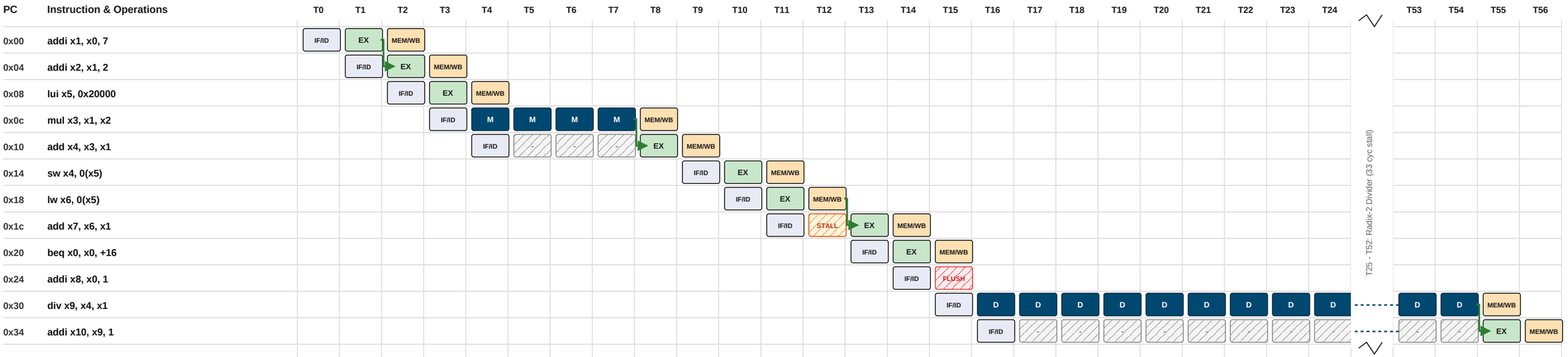
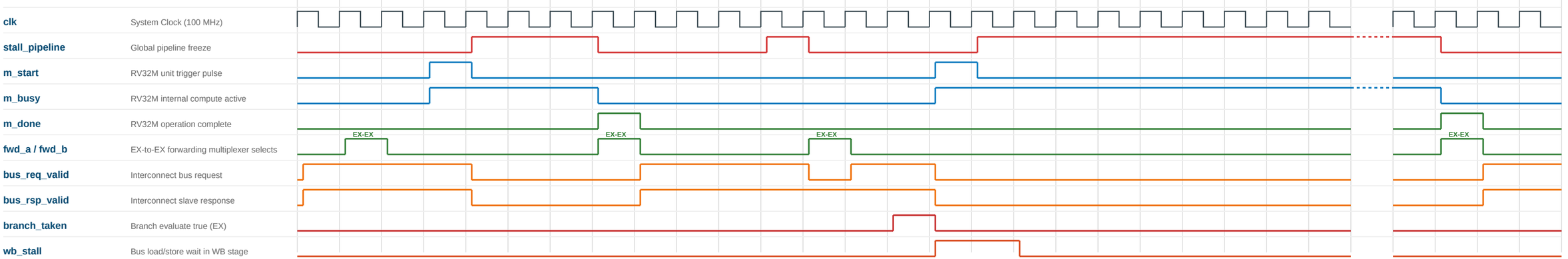
Figure 3 - Pipeline Timing: EX-to-EX Forwarding, Load-ALU Stall, Bus Wait & RV32M Multi-Cycle Ops

Exact cycle-by-cycle trace from cpu_core + sys_bus_interconnect + imem/dmem slaves | 1 Cycle = 1 Clock Period (10 ns @ 100 MHz) | Stages: IF/ID (Fetch+Decode), EX (Execute/M-Unit), MEM/WB (Bus/Writeback) | tcq propagation delay shown on stages & signals



HARDWARE CONTROL SIGNALS & BUS PROTOCOL (RTL cycle-accurate waveforms from sim)



Stage Key: IF/ID (Fetch/Decode) | EX (Single-cycle ALU) | M / D (Multiply / Divide Multi-Cycle EX) | MEM/WB (Memory/Writeback) | Load-ALU Stall (Bubble) | Pipeline Freeze / Bus Hold | Flushed (Branch Squash)

Pipeline Hazards & Forwarding Details:

- EX-to-EX Forwarding (Green Arrows): T2 (addi x1 -> addi x2), T8 (mul x3 -> add x4), T13 (lw x6 -> add x7), T55 (div x9 -> addi x10). Eliminates RAW bubbles.
- Multi-Cycle & Control: MUL takes 4 cycles in EX (T4-T7). Load-ALU Stall at T12 (1 cycle). Branch taken at T14 squashes single wrong +4 fetch (0x24 at T15); target 0x30 is fetched in T15 while branch is in WB. DIV takes 34 cycles. tcq delay depicted on all transitions.